Library Service GBV

NORMAL

Belarusian State University's Library

4 Nezavisimosti av.
220030 Minsk

Herr/Frau
Belarusian State University's LibraryBelarusian State University's Library
SLI04X00454

Bestellende Bibliothek: <SEV.SUBITO.ORDER>
Belarusian State University's Library

4 Nezavisimosti av.
220030 Minsk
edd@bsu.by (+375-17)209-50-37
USER-GROUP-8

Benutzer-Ausweisnummer
SLI04X00454

Verfasser: Han, Qing
(Aufsatz)
Titel: Distributions and probability generating functions of success run vector
(Aufsatz)
Seiten: 57-72
Titel:
(Monographie oder Zeitschrift)
Far East journal of theoretical statistics

Allahabad

0972-0863

Standort:
ZA 88463 / FMAG

Ablage
6,1 92

Band/Heft VOL 6; NUM 1; 57-72
Jahrgang 2002

Lieferform PDF
Lieferart EMAIL

PPN:
Bemerkung:

Lieferschein

Abrechnung über Subito-Zentralregulierung

Lieferbibliothek: <7> SUB Göttingen, GAUSS

Bearbeitungsdatum: 12. Jan. 2006

☐ Eildienst (24 Std/ hrs) ☐ Altbestand

Anzahl Kopien: 16

☐ Kopie/ hardcopy ☒ E-Mail ☐ Ariel ☐ Fax

☐ Ausleihe eines Buches (Mikroform)/lending of book (microform)

Hausadresse:
Niedersächsische Staats- und Universitätsbibliothek Göttingen,
Fernleihe, Platz der Göttinger Sieben 1,
37070 Göttingen
GERMANY

Telefon: +49(0)551/39-2035, -5276
Fax: +49(0)551/39-5014
E-Mail: gauss@mail.sub.uni-goettingen.de
Ust-IdNr: DE152336201

Far East J. Theo. Stat. 6(1) (2002), 57-72

DISTRIBUTIONS AND PROBABILITY GENERATING FUNCTIONS OF SUCCESS RUN VECTOR

QING HAN

The Institute of Statistical Mathematics
Tokyo 106-8569, Japan

and

Department of Statistics
East China Normal University
Shanghai 200062, P. R. China

Abstract

In this paper, we consider the number of success runs of length k_1 in the first component and the number of success runs of length k_2 in the second component in Markov dependent bivariate $\{0, 1\}^2$-valued trials. By using a Markov chain imbedding method, we obtain the joint distributions and the probability generating functions of the success run vector.

1. Introduction

In recent years, discrete distribution theory related to run statistics has been developed by many authors. One of the reasons for this, is their widespread application, for example, the reliability of consecutive-k-out-of-n: F systems (Chao et al. [4]), start-up demonstration test (Balakrishnan et al. [2]) and DNA sequence analysis (Doi and Yamamoto [5] and Ewens and Grant [6]).

2000 Mathematics Subject Classification: 60E05, 62E15.

Key words and phrases: success run, Markov dependent trials, Markov chain imbedding method, probability generating function, bivariate trials.

Received December 6, 2001

There are various definitions of run. The four most important and frequently used success runs are:

(a) $E_{n,\,k}$, the number of success runs of size exactly k until the n-th trial (Mood [18]);

(b) $N_{n,\,k}$, the number of nonoverlapping consecutive k successes until the n-th trial (Feller [7]);

(c) $M_{n,\,k}$, the number of overlapping consecutive k successes until the n-th trial (Ling [17]);

(d) $G_{n,\,k}$, the number of success runs of size greater than or equal to k until the n-th trial.

The probability functions $(p.f.)$ and the probability generating functions $(p.g.f.)$ of these run statistics have been studied by many authors in many situations. Fu and Koutras [8] and Koutras and Alexandrou [15] studied the distributions of run statistics based on a Markov chain imbedding technique. Fu [9] and Han and Aki [12] discussed the joint distributions of run in a sequence of multi-state trials. Godbole et al. [10] and Han and Aki [11] derived the joint distributions of runs of several lengths in Bernoulli trials and a two-state Markov chain. Koutras and Alexandrou [16] discussed the waiting time problems of run in a sequence of trinomial trials. Han and Aki [13] obtained the distribution of sooner and later waiting time of several runs. Aki and Hirano [1] studied the waiting time problems of run in Markov dependent bivariate trials.

In this paper, we introduced a Markov chain imbeddable vector of a new type to study the joint distributions of the most common run statistics $(E_{n,\,k},\, N_{n,\,k},\, M_{n,\,k}$ and $G_{n,\,k})$ in Markov dependent bivariate trials. A number of well known results can be obtained as special cases of the formulas stated here. These results are also suitable for numerical and symbolic calculations. The method can be extended to a sequence of multivariate trials and a sequence of multi-state multivariate trials.

The results in this paper can be applied to the statistical analyses of reliability of the consecutive-k-out-of-n: F system. For consecutive-k-out-of-n: F system, we refer the reader Chang et al. [3]. For example, we can mention a microwave station system. Each microwave station is able to transmit signal A (variable X) a distance up to k_1 microwave stations and transmit signal B (variable Y) a distance up to k_2 microwave stations. This system fails in transmitting signal A iff at least k_1 consecutive microwave stations fail and fails of transmitting signal B iff at least k_2 consecutive microwave stations fail. About other applications, we refer to Aki and Hirano [1].

In Section 2, we introduce a Markov chain imbeddable vector of non-jumping type and obtain some main results about the $p.f.$ and the $p.g.f.$ of the Markov chain imbeddable vector. In Section 3, we obtain the joint distributions of success run vector in Markov dependent bivariate trials, by using the results in Section 2.

2. Main Results

Let $\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix}, \begin{pmatrix} X_2 \\ Y_2 \end{pmatrix}, \ldots,$ be a sequence of $\left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$-valued bivariate trials with the following probabilities: $\Pr\left(\begin{pmatrix} X_t \\ Y_t \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}\right) = p_{xy,\,t}$, $x, y = 0, 1$ and $t = 1, 2, \ldots,$ where $p_{00,\,t} + p_{01,\,t} + p_{10,\,t} + p_{11,\,t} = 1,$ for all $t = 1, 2, \ldots.$

Let $\mathbf{R_t} = \begin{pmatrix} R_{1t} \\ R_{2t} \end{pmatrix}$, where R_{1t} is the number of 1-run of length k_1 in $X_1, \ldots, X_t$, and R_{2t} is the number of 1-run of length k_2 in $Y_1, \ldots, Y_t$. We consider the probability function and the probability generating function of $\mathbf{R}_n$.

First, we introduce some notations. Let $\mathbf{r} = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$ be an $\mathbf{R}_n$'s realization. Let $\mathbf{l}_n = \begin{pmatrix} l_{1n} \\ l_{2n} \end{pmatrix}$, where $l_{1n} = \max\{r_1 : \Pr(R_{1n} = r_1) > 0\}$ and

$l_{2n} = \max\{r_2 : \Pr(R_{2n} = r_2) > 0\}$. Hence, $\Pr(\mathbf{R}_n = \mathbf{r}) = 0$, for all $\mathbf{r} \notin \{\mathbf{r} : \mathbf{0} \le \mathbf{r} \le \mathbf{l}_n\}$. Let $\mathbf{1} = (1, 1, ..., 1)$.

Definition 2.1. A random vector $\mathbf{R}_n$ will be called a *Markov chain imbeddable vector* of non-jumping type, if

(1) there exists a Markov chain $\{Z_t : t \ge 0\}$ defined on a state space Ω,

(2) there exists a partition $\{\mathbf{C}_\mathbf{r}, \mathbf{r} \ge \mathbf{0}\}$ on the state space Ω,

(3) for every $\mathbf{r}$, $\Pr(\mathbf{R}_n = \mathbf{r}) = \Pr(Z_n \in \mathbf{C}_\mathbf{r})$,

(4) $\Pr\left(Z_t \in \mathbf{C}_{\mathbf{r}+\binom{i}{j}} \mid Z_{t-1} \in \mathbf{C}_\mathbf{r} \right) = 0$, for all i or $j \ne -1, 0, 1$.

From (4), follows that the Markov chain cannot jump directly to a higher (or lower) state without visiting one of neighboring state $\mathbf{C}_{\mathbf{r}+\binom{i}{j}}$, $(i, j = -1, 0, 1)$. So, we use the nomenclature *non-jumping type.*

Without loss of generality, we can assume $\mathbf{C}_\mathbf{r}$ has the common cardinality $s = |\mathbf{C}_\mathbf{r}|$ for every $\mathbf{r}$, and denote $\mathbf{C}_\mathbf{r} = \{C_{\mathbf{r},1}, C_{\mathbf{r},2}, ..., C_{\mathbf{r},s}\}$.

We introduce the probability vectors of the t-th step Z_t of the Markov chain

$$\mathbf{f}_t(\mathbf{r}) = (\Pr(Z_t = C_{\mathbf{r},1}), ..., \Pr(Z_t = C_{\mathbf{r},s})), \ \mathbf{0} \le \mathbf{r} \le \mathbf{l}_t, \ (t = 0, 1, ..., n),$$

where $\mathbf{l}_t = \binom{l_{1t}}{l_{2t}}$ and l_{it} $(i = 1, 2)$ is the largest i-th element in subscripts $\mathbf{r}$ of partition $\mathbf{C}_\mathbf{r}$ which include accessible state of the t-th step Z_t of the Markov chain, i.e., $l_{1t} = \max\left\{ r_1 : \Pr\left(Z_t \in \mathbf{C}_{\binom{r_1}{\bullet}} \right) > 0 \right\}$. When $t = n$, it is $\mathbf{l}_n$.

$\mathbf{f}_0(\mathbf{r})$ is the initial probabilities of the Markov chain $\{Z_t, t \ge 0\}$, i.e.,

$\mathbf{f}_0(\mathbf{r}) = (\Pr(Z_0 = C_{\mathbf{r},1}), \ldots, \Pr(Z_0 = C_{\mathbf{r},s}))$, $\mathbf{0} \le \mathbf{r} \le \mathbf{l}_0$. In the applications for run statistics, we have $\mathbf{l}_0 = \mathbf{0}$.

We introduce the transition probability matrices

$$T_t^{(k,l)}(\mathbf{r}) = \left(\Pr\left(Z_t = C_{\mathbf{r}+\binom{k}{l}, j} \mid Z_{t-1} = C_{\mathbf{r}, i}\right)\right)_{s\times s},$$

$$(k, l = -1, 0, 1), (t = 1, \ldots, n).$$

Clearly, the entries of $T_t^{(k,l)}(\mathbf{r})$ contrail between the states $\mathbf{C}_\mathbf{r}$ and $\mathbf{C}_{\mathbf{r}+\binom{k}{l}}$ one-step transitions $(k, l = -1, 0, 1)$.

Using the Chapman-Kolmogorav equations, Definition 2.1 and the form of the transition probability matrices, we have

$$\mathbf{f}_t(\mathbf{r}) = \sum_{i=-1}^{1}\sum_{j=-1}^{1} \mathbf{f}_{t-1}\left(\mathbf{r} - \binom{i}{j}\right) T_t^{(i,j)}\left(\mathbf{r} - \binom{i}{j}\right), \quad (\mathbf{0} \le \mathbf{r} \le \mathbf{l}_t), (t = 1, 2, \ldots, n),$$

where $\mathbf{f}_t(\mathbf{r}) = \mathbf{0}$, for all $\mathbf{r} \notin \{\mathbf{r} : \mathbf{0} \le \mathbf{r} \le \mathbf{l}_t\}$, $(t = 1, \ldots, n)$. And, the probability function of $\mathbf{R}_n$ is given by

$$\Pr(\mathbf{R}_n = \mathbf{r}) = \Pr(Z_n \in \mathbf{C}_\mathbf{r}) = \sum_{i=1}^{s} \Pr(Z_n = C_{\mathbf{r}, i}) = \mathbf{f}_n(\mathbf{r}) \cdot \mathbf{1}'.$$

Using above formulas, we can evaluate the joint distributions of run statistics. But, the evaluation is complicated. Noting these matrices $T_t^{(i,j)}(\mathbf{r})\,(i, j = -1, 0, 1)$ usually do not depend on $\mathbf{r}$ in the distribution problem of run, in the following paper, we discuss the special case.

We consider the probability generating function of $\mathbf{R}_n$,

$$\psi_n(u, v) = \sum_{\mathbf{0}\le\mathbf{r}\le\mathbf{l}_n} \Pr(\mathbf{R}_n = \mathbf{r}) u^{r_1} v^{r_2}.$$

Defining $\phi_t(u, v) = \sum_{\mathbf{0}\le\mathbf{r}\le\mathbf{l}_t} \mathbf{f}_t(\mathbf{r}) u^{r_1} v^{r_2}$, we have $\psi_n(u, v) = \phi_n(u, v)\mathbf{1}'$.

Theorem 2.1. *If* $T_t^{(i,\,j)}(\mathbf{r})$, $(i,\ j = -1,\ 0,\ 1)$ *do not depend on* $\mathbf{r}$, *i.e.*, $T_t^{(i,\,j)}(\mathbf{r}) = T_t^{(i,\,j)}$, $(i,\ j = -1,\ 0,\ 1)$, *for all* $\mathbf{r}$, *then*

$$\phi_t(u,\ v) = \phi_{t-1}(u,\ v)\left(\sum_{i=-1}^{1}\sum_{j=-1}^{1} T_t^{(i,\,j)} u^i v^j\right).$$

Hence, the probability generating function of $\mathbf{R}_n$ *is*

$$\psi_n(u,\ v) = \phi_0(u,\ v)\prod_{t=1}^{n}\left(\sum_{i=-1}^{1}\sum_{j=-1}^{1} T_t^{(i,\,j)} u^i v^j\right)\mathbf{1}',$$

where $\phi_0(u,\ v) = \sum_{\mathbf{0}\le\mathbf{r}\le\mathbf{l}_0} \mathbf{f}_0(\mathbf{r}) u^{r_1} v^{r_2}$.

Proof. See Appendix.

It is worth mentioning that in most of applications of runs, we have $\mathbf{l}_0 = \mathbf{0}$ and $\mathbf{f}_0(\mathbf{0}) = (1,\ 0,\ \ldots,\ 0)$. In this case, we have $\phi_0(u,\ v) = (1,\ 0,\ \ldots,\ 0)$.

Remark 2.1. Let $v = 1$, and

$$A_t = \sum_{j=-1}^{1} T_t^{(0,\,j)} = \left(\Pr\left(Z_t \in \cup_{j=-1}^{1} \mathbf{C}_{\mathbf{r}+\binom{0}{j}} \mid Z_{t-1} \in \mathbf{C}_\mathbf{r}\right)\right),$$

$$B_t = \sum_{j=-1}^{1} T_t^{(1,\,j)}$$

and

$$C_t = \sum_{j=-1}^{1} T_t^{(-1,\,j)}.$$

Then the marginal probability generating function of R_{1t} is given by

$$\phi_t(u) = \phi_{t-1}(u)(A_t + B_t u + C_t u^{-1}).$$

This is Theorem 4.2 of Han and Aki [12].

2.2. Assuming $T^{(i,\,j)} = \mathbf{0}$, for all $i = -1$ or $j = -1$, i.e., the Markov chain $\{Z_t,\ t \ge 0\}$ cannot move backwards, we have

$$\phi_t(u, v) = \phi_{t-1}(u, v)(T^{(0, 0)} + T^{(0, 1)}v + T^{(1, 0)}v + T^{(1, 1)}uv).$$

This is Theorem 2.2 of Han [14].

Further, for $v = 1$, $\phi_t(u) = \phi_{t-1}(u)[(T^{(0, 0)} + T^{(0, 1)} + T^{(1, 0)}) + T^{(1, 1)}u]$. It is the result of Koutras and Alexandrou [15].

Theorem 2.2. *If $T_t^{(i, j)}(\mathbf{r}) = T^{(i, j)}$, $(i, j = -1, 0, 1)$, for all t and $\mathbf{r}$, and $\mathbf{l}_0 = \mathbf{0}$, then*

$$E(R_{1t}) = \phi_0(\mathbf{1}) \sum_{k=1}^{n} A^{k-1} B \mathbf{1}',$$

$$E(R_{2t}) = \phi_0(\mathbf{1}) \sum_{k=1}^{n} A^{k-1} C \mathbf{1}',$$

$$E(R_{1t} R_{2t}) = \phi_0(\mathbf{1}) \sum_{k=1}^{n} \left\{ \sum_{l=1}^{k-1} A^{l-1} B A^{k-1-l} C + A^{k-1} D + A^{k-1} C \sum_{l=1}^{n-k} A^{l-1} B \right\} \mathbf{1}',$$

where $A = \sum_{i=-1}^{1} \sum_{j=-1}^{1} T^{(i, j)}$, $B = \sum_{j=-1}^{1} T^{(1, j)} - \sum_{j=-1}^{1} T^{(-1, j)}$, $C = \sum_{i=-1}^{1} T^{(i, 1)} - \sum_{i=-1}^{1} T^{(i, -1)}$ *and* $D = T^{(1, 1)} + T^{(-1, -1)} - T^{(-1, 1)} - T^{(1, -1)}$.

Proof. Taking the derivative in $\psi_n(u, v)$ with respect to u or v and setting $(u, v) = (1, 1)$, and using $\left(\sum_{i=-1}^{1} \sum_{j=-1}^{1} T^{(i, j)}\right) \cdot \mathbf{1}' = \mathbf{1}'$ and

$$\frac{d}{dz}(A + zB + z^{-1}C)^n$$

$$= \sum_{k=1}^{n} (A + zB + z^{-1}C)^{k-1} (B - z^{-2}C)(A + zB + z^{-1}C)^{n-k},$$

we can obtain these results.

3. The Joint Distributions of Success Run Vector

In this section, using the results of Section 2, we obtain the joint distributions of success run vector in Markov dependent bivariate trials.

Let $\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix}, \begin{pmatrix} X_2 \\ Y_2 \end{pmatrix}, \dots, \begin{pmatrix} X_n \\ Y_n \end{pmatrix}$ be a sequence of $\left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$-valued Markov chain, with the following probabilities: for $x, y, w, z = 0, 1$ and $t = 2, \dots, n$, $p_{xy} = \Pr\left(\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}\right)$ and $p(x, y | w, z) = \Pr\left(\begin{pmatrix} X_t \\ Y_t \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \middle| \begin{pmatrix} X_{t-1} \\ Y_{t-1} \end{pmatrix} = \begin{pmatrix} w \\ z \end{pmatrix}\right)$, where $p_{00} + p_{01} + p_{10} + p_{11} = 1$ and $\sum_{x=0}^{1} \sum_{y=0}^{1} p(x, y | w, z) = 1$ for every $w, z = 0, 1$.

Let us denote by $r_1 (r_2)$ the number of "1"-runs of length $k_1 (k_2)$ up to the t-th trial in $X_1, \dots, X_t$ $(Y_1, \dots, Y_t)$. Let $r_1^* (r_2^*)$ be the number of trailing "1" in $X_1, \dots, X_t$ $(Y_1, \dots, Y_t)$, i.e., the number of last consecutive "1" counting backwards in $X_1, \dots, X_t$ $(Y_1, \dots, Y_t)$.

We consider $\mathbf{R}_n = \begin{pmatrix} E_{n, k_1} \\ N_{n, k_2} \end{pmatrix}$ in the sequence of $\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix}, \begin{pmatrix} X_2 \\ Y_2 \end{pmatrix}, \dots, \begin{pmatrix} X_n \\ Y_n \end{pmatrix}$.

From definitions of E_{n, k_1} and N_{n, k_2}, we have $l_{1n} = \left[\dfrac{n+1}{k_1+1}\right]$, $l_{2n} = \left[\dfrac{n}{k_2}\right]$. Let

$$i_1 = \begin{cases} r_1^*, & \text{if } r_1^* \le k - 1 \\ -1, & \text{if } r_1^* = k \\ -2, & \text{if } r_1^* > k, \end{cases}$$

$$i_2 = \begin{cases} r_2^* - \left[\dfrac{r_2^*}{k_2}\right] k_2, & \text{if } \left[\dfrac{r_2^*}{k_2}\right] \notin \{1, 2, \dots\} \\ -1, & \text{if } \left[\dfrac{r_2^*}{k_2}\right] \in \{1, 2, \dots\}. \end{cases}$$

We can give a Markov chain $\{Z_t = (r_1, r_2; i_1, i_2) | t \geq 0\}$. The state space is

$$\Omega = \{(r_1, r_2; i_1, i_2) | 0 \leq r_1 \leq l_{1n}; 0 \leq r_2 \leq l_{2n};$$

$$i_1 = 0, 1, \dots, k_1 - 1, -1, -2; i_2 = 0, 1, \dots, k_2 - 1, -1\}.$$

Define $\mathbf{C}_{(r_1, r_2)} = \{(r_1, r_2; i_1, i_2) | i_1 = 0, 1, \dots, k_1 - 1, -1, -2; i_2 = 0, 1, \dots, k_2 - 1, -1\}$, $r_i = 0, 1, \dots, l_{in}$ $(i = 1, 2)$. Then

$$\Omega = \bigcup_{r_1=0}^{l_{1n}} \bigcup_{r_2=0}^{l_{2n}} \mathbf{C}_{(r_1, r_2)}.$$

Using lexicographical order, we let $\mathbf{C}_{(r_1, r_2)} = \{(r_1, r_2; 0, 0), (; 0, 1), \dots, (; 0, k_2 - 1), (; 0, -1), (; 1, 0), (; 1, 1), \dots, (; 1, k_2 - 1), (; 1, -1), \dots, (; k_1 - 1, 0), \dots, (; k_1 - 1, k_2 - 1), (; k_1 - 1, -1), (; -1, 0), \dots, (; -1, k_2 - 1), (; -1, -1), (; -2, 0), \dots, (; -2, k_2 - 1), (; -2, -1)\} \equiv \{C_{(x, y), 1}, C_{(x, y), 2}, \dots, C_{(x, y), s}\}$, where $s = (k_1 + 2) \times (k_2 + 1)$.

With this set up, the random vector $\mathbf{R}_n = \begin{pmatrix} E_{n, k_1} \\ N_{n, k_2} \end{pmatrix}$ satisfies

Definition 2.1. From $Z_{t-1} = (r_1, r_2; i_1, i_2)$, we obtain the value of $\begin{pmatrix} X_{t-1} \\ Y_{t-1} \end{pmatrix}$, $(t = 1, \dots, n)$, as

$$\begin{pmatrix} X_{t-1} \\ Y_{t-1} \end{pmatrix} = \begin{cases} \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & \text{if } i_1 = 0, i_2 = 0 \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, & \text{if } i_1 = 0, i_2 \neq 0 \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, & \text{if } i_1 \neq 0, i_2 = 0 \\ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, & \text{if } i_1 \neq 0, i_2 \neq 0. \end{cases}$$

Hence, we can get transition probabilities from Z_{t-1} to Z_t. And, these transition probability matrices $T_t^{(i, j)}$, $(i, j = -1, 0, 1)$, $(t = 1, \dots, n)$ can be obtained.

As an illustration, let $k_1 = 2$, $k_2 = 2$, then we have $s = 12$ and $\mathbf{C}_{(r_1, r_2)} = \{(r_1, r_2; 0, 0), (; 0, 1), (; 0, -1), (; 1, 0), (; 1, 1), (; 1, -1), (; -1, 0), (; -1, 1), (; -1, -1), (; -2, 0), (; -2, 1), (; -2, -1)\}$, with

$$T_t = \sum_{i=-1}^{1} \sum_{j=-1}^{1} T_t^{(i,\, j)} u^i v^j =$$

$$\begin{pmatrix}
p_{0|0} & p_{1|0} & 0 & p_{2|0} & p_{3|0} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
p_{0|1} & 0 & p_{1|1}v & p_{2|1} & 0 & p_{3|1} & 0 & 0 & 0 & 0 & 0 & 0 \\
p_{0|1} & p_{1|1} & 0 & p_{2|1} & p_{3|1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
p_{0|2} & p_{1|2} & 0 & 0 & 0 & 0 & p_{2|2}u & p_{3|2}u & 0 & 0 & 0 & 0 \\
p_{0|3} & 0 & p_{1|3}v & 0 & 0 & 0 & p_{2|3}u & 0 & p_{3|3}uv & 0 & 0 & 0 \\
p_{0|3} & p_{1|3} & 0 & 0 & 0 & 0 & p_{2|3}u & p_{3|3}u & 0 & 0 & 0 & 0 \\
p_{0|2} & p_{1|2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & p_{2|2}u^{-1} & p_{3|2}u^{-1} & 0 \\
p_{0|3} & 0 & p_{1|3}v & 0 & 0 & 0 & 0 & 0 & 0 & p_{2|3}u^{-1} & 0 & p_{3|3}u^{-1}v \\
p_{0|3} & p_{1|3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & p_{2|3}u^{-1} & p_{3|3}u^{-1} & 0 \\
p_{0|2} & p_{1|2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & p_{2|2} & p_{3|2} & 0 \\
p_{0|3} & 0 & p_{1|3}v & 0 & 0 & 0 & 0 & 0 & 0 & p_{2|3} & 0 & p_{3|3} \\
p_{0|3} & p_{1|3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & p_{2|3} & p_{3|3} & 0
\end{pmatrix}$$

$$(t = 2, \ldots, n),$$

where $p_{2x+y|2w+z} = p(x, y | w, z)$, $x, y, w, z = 0, 1$. In the above matrix, the "0" at these subscripts denote the outcome is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, the "1" denotes $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, the "2" denotes $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and the "3" denotes $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

The matrix $T_1 = \sum_{i=-1}^{1} \sum_{j=-1}^{1} T_1^{(i,\, j)} u^i v^j$ is different from T_t, $(t = 2, \ldots, n)$. The difference is only that the first row of $T_1 = \sum_{i=-1}^{1} \sum_{j=-1}^{1} T_1^{(i,\, j)} u^i v^j$ is $(p_{00}, p_{01}, 0, p_{10}, p_{11}, 0, 0, 0, 0, 0, 0, 0)$.

Using the results of Section 2, we can get the probability generating function of $\mathbf{R}_n$,

$$\psi_n(u, v) = \phi_0(u, v) T_1 \cdot T_2^{n-1} \mathbf{1}',$$

where $\phi_0(u, v) = (1, 0, \ldots, 0)$. From the *p.g.f.*, we can derive the probability function of $\mathbf{R}_n$.

Remark 3.1. In i.i.d. case, we have $T_1 = T_2 = \cdots = T_n$, Theorem 2.2 can be used.

3.2. In Markov dependent sequence, conditions of Theorem 2.2 are not satisfied. But, some trivial modifications are enough to get a similar result.

3.3. In the example above, our state space about $N_{n,\,k}$-run is different from Koutras and Alexandrou's [15]. In their paper, the state $(\cdot;\, 0)$ denote two cases: (a) just occurring "Failure" or (b) occurring "Success" and just occurring consecutive k successes. So, it is not suitable in Markov dependent sequence. The matrices $A_t(x)$ and $B_t(x)$ are wrong in Section 7 of Koutras and Alexandrou [15, p. 763].

As in Han and Aki [12], we discuss the overlapping runs (M-runs) and the runs of lengths at least $(k_1,\, k_2)$ (G-runs).

For $M_{n,\,k}$ run, $l_n = n - k + 1$, and

$$i = \begin{cases} r^*, & \text{if } r^* \le k - 1 \\ -1, & \text{if } r^* \ge k. \end{cases}$$

For $G_{n,k}$ run, $l_n = \left[\dfrac{n+1}{k+1}\right]$ and

$$i = \begin{cases} r^*, & \text{if } r^* \le k - 1 \\ -1, & \text{if } r^* \ge k. \end{cases}$$

An advantage of our method is that the dimension of the transition probability matrices T_t is independent of the number n of trials. The evaluation of the joint distributions of run vector is easily performed recursively at large sequences of trials. Obviously, our method can also be extended to a sequence of multivariate trial and a sequence of multi-state bivariate trials.

Appendix

Proof of Theorem 2.1.

$$\phi_t(u, v) = \sum_{0 \le \mathbf{r} \le \mathbf{l}_t} \mathbf{f}_t(r_1, r_2) u^{r_1} v^{r_2}$$

$$= \sum_{r_1=0}^{l_{1t}} \sum_{r_2=0}^{l_{2t}} \left[\sum_{i=-1}^{1} \sum_{j=-1}^{1} \mathbf{f}_{t-1}(r_1 - i, r_2 - j) T_t^{(i, j)} \right] u^{r_1} v^{r_2}$$

$$= \sum_{i=-1}^{1} \sum_{j=-1}^{1} \left[\sum_{r_1=-i}^{l_{1t}-i} \sum_{r_2=-j}^{l_{2t}-j} \mathbf{f}_{t-1}(r_1, r_2) T_{t-1}^{(i, j)} u^{r_1} v^{r_2} \right] u^i v^j,$$

$$(t = 1, 2, \ldots, n).$$

For convenience, let

$$\mathbf{g}_t(u, v, i, j) = \left[\sum_{r_1=-i}^{l_{1t}-i} \sum_{r_2=-j}^{l_{2t}-j} \mathbf{f}_{t-1}(r_1, r_2) T_{t-1}^{(i, j)} u^{r_1} v^{r_2} \right],$$

$$(i, j = -1, 0, 1), (t = 1, 2, \ldots, n).$$

Noting that $\mathbf{R}_n$ is a Markov chain imbeddable vector of non-jumping type, we have four cases: (a) $\mathbf{l}_t = \mathbf{l}_{t-1}$, (b) $\mathbf{l}_t - \mathbf{l}_{t-1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, (c) $\mathbf{l}_t - \mathbf{l}_{t-1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and (d) $\mathbf{l}_t - \mathbf{l}_{t-1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

And, $\mathbf{f}_t(\mathbf{r}) = \mathbf{0}$ for all $\mathbf{r} \notin \{\mathbf{r} : \mathbf{0} \le \mathbf{r} \le \mathbf{l}_t\}$, $(t = 1, 2, \ldots, n)$. We have

$$\mathbf{g}_t(u, v, -1, -1) = \sum_{r_1=1}^{l_{1t}+1} \sum_{r_2=1}^{l_{2t}+1} \mathbf{f}_{t-1}(r_1, r_2) u^{r_1} v^{r_2} T_t^{(-1, -1)}$$

$$= \left[\sum_{r_1=0}^{l_{1, t-1}} \sum_{r_1=0}^{l_{2, t-1}} - \sum_{r_1=0}^{l_{1, t-1}} \sum_{r_2=0}^{0} - \sum_{r_2=1}^{l_{2, t-1}} \sum_{r_1=0}^{0} \right] \mathbf{f}_{t-1}(r_1, r_2) u^{r_1} v^{r_2} T_t^{(-1, -1)}$$

$$= \phi_{t-1}(u, v) T_t^{(-1, -1)} - \sum_{r_1=0}^{l_{1, t-1}} \mathbf{f}_{t-1}(r_1, 0) u^{r_1} T_t^{(-1, -1)}$$

$$-\sum_{r_2=1}^{l_{2,t-1}} \mathbf{f}_{t-1}(0, r_2) v^{r_2} T_t^{(-1,-1)}.$$

Similarly, we have

$$\mathbf{g}_t(u, v, -1, 0) = \phi_{t-1}(u, v) T_t^{(-1,0)} - \sum_{r_2=0}^{l_{2,t-1}} \mathbf{f}_{t-1}(0, r_2) v^{r_2} T_t^{(-1,0)},$$

$$\mathbf{g}_t(u, v, 0, -1) = \phi_{t-1}(u, v) T_t^{(0,-1)} - \sum_{r_1=0}^{l_{1,t-1}} \mathbf{f}_{t-1}(r_1, 0) u^{r_1} T_t^{(0,-1)},$$

$$\mathbf{g}_t(u, v, 0, 0) = \phi_{t-1}(u, v) T_t^{(0,0)}.$$

$$\mathbf{g}_t(u, v, -1, 1) = \left[\sum_{r_1=0}^{l_{1,t-1}} \sum_{r_2=0}^{l_{2t}-1} - \sum_{r_2=0}^{l_{2t}-1} \sum_{r_1=0}^{0}\right] \mathbf{f}_{t-1}(r_1, r_2) u^{r_1} v^{r_2} T_t^{(-1,1)}$$

$$\text{in (a, c) cases:} = \left[\sum_{r_1=0}^{l_{1,t-1}} \sum_{r_2=0}^{l_{2,t-1}-1} - \sum_{r_2=0}^{l_{2,t-1}-1} \sum_{r_1=0}^{0}\right] \mathbf{f}_{t-1}(r_1, r_2) u^{r_1} v^{r_2} T_t^{(-1,1)}$$

$$= \left[\sum_{r_1=0}^{l_{1,t-1}} \sum_{r_2=0}^{l_{2,t-1}} - \sum_{r_1=0}^{l_{1,t-1}} \sum_{r_2=l_{2,t-1}}^{l_{2,t-1}} - \sum_{r_2=0}^{l_{2,t-1}-1} \sum_{r_1=0}^{0}\right] \mathbf{f}_{t-1}(r_1, r_2) u^{r_1} v^{r_2} T_t^{(-1,1)}$$

$$= \phi_{t-1}(u, v) T_t^{(-1,1)} - \sum_{r_1=0}^{l_{1,t-1}} \mathbf{f}_{t-1}(r_1, l_{2,t-1}) u^{r_1} v^{l_{2,t-1}} T_t^{(-1,1)} - \sum_{r_2=0}^{l_{2,t-1}} f_{t-1}(0, r_2) v^{r_2} T_t^{(-1,1)};$$

$$\text{in (b, d) cases:} = \phi_{t-1}(u, v) T_t^{(-1,1)} - \sum_{r_2=0}^{l_{2,t-1}} \mathbf{f}_{t-1}(0, r_2) v^{r_2} T_t^{(-1,1)},$$

$\mathbf{g}_t(u, v, 1, -1) =$

$$\text{in (c, d) cases: } = \phi_{t-1}(u, v)T_t^{(1, -1)} - \sum_{r_1=0}^{l_{1, t-1}} \mathbf{f}_{t-1}(r_1, 0)u^{r_1}T_t^{(1, 0)};$$

$$\text{in (a, b) cases: } = \phi_{t-1}(u, v)T_t^{(1, -1)} - \sum_{r_2=0}^{l_{2, t-1}} \mathbf{f}_{t-1}(l_{1, t-1}, r_2)u^{l_{1, t-1}}v^{r_2}T_t^{(1, -1)} - \sum_{r_1=0}^{l_{1, t-1}} \mathbf{f}_{t-1}(r_1, 0)u^{r_1}T_t^{(1, -1)},$$

$\mathbf{g}_t(u, v, 0, 1) =$

$$\text{in (a, c) cases: } = \phi_{t-1}(u, v)T_t^{(0, 1)} - \sum_{r_1=0}^{l_{1, t-1}} \mathbf{f}_{t-1}(r_1, l_{2, t-1})u^{r_1}v^{l_{2, t-1}}T_t^{(0, 1)};$$

$$\text{in (b, d) cases: } = \phi_{t-1}(u, v)T_t^{(0, 1)},$$

$\mathbf{g}_t(u, v, 1, 0) =$

$$\text{in (c, d) cases: } = \phi_{t-1}(u, v)T_t^{(1, 0)};$$

$$\text{in (a, b) cases: } = \phi_{t-1}(u, v)T_t^{(1, 0)} - \sum_{r_2=0}^{l_{2, t-1}} \mathbf{f}_{t-1}(l_{1, t-1}, r_2)u^{l_{1, t-1}}v^{r_2}T_t^{(1, 0)},$$

$\mathbf{g}_t(u, v, 1, 1) =$

$$\text{in (a) case: } = \phi_{t-1}(u, v)T_t^{(1, 1)} - \sum_{r_1=0}^{l_{1, t-1}} \mathbf{f}_{t-1}(r_1, l_{2, t-1})u^{r_1}v^{l_{2, t-1}}T_t^{(1, 1)} - \sum_{r_2=0}^{l_{2, t-1}-1} \mathbf{f}_{t-1}(l_{1, t-1}, r_2)u^{l_{1, t-1}}v^{r_2};$$

$$\text{in (b) case: } = \phi_{t-1}(u, v)T_t^{(1,1)} - \sum_{r_2=0}^{l_{2, t-1}} \mathbf{f}_{t-1}(l_{1,t-1}, r_2)u^{l_{1, t-1}}v^{r_2}T_t^{(1,1)};$$

in (c) case: $= \phi_{t-1}(u, v) T_t^{(1, 1)} - \sum_{r_1=0}^{l_{1, t-1}} \mathbf{f}_{t-1}(r_1, l_{2, t-1}) u^{r_1} v^{l_{2, t-1}} T_t^{(1, 1)}$;

in (d) case: $= \phi_{t-1}(u, v) T_t^{(1, 1)}$.

Hence,

$$\phi_t(u, v) = \sum_{i=-1}^{1} \sum_{j=-1}^{1} \phi_{t-1}(u, v) T_t^{(i, j)} u^i v^j - \mathbf{E}, \quad (t = 1, 2, \ldots, n).$$

Let $u = v = 1$ and multiply both sides by $\mathbf{1}$. Because $\phi_t(1, 1)\mathbf{1}' = \psi_t(1, 1) = 1$, and $\left(\sum_{i=-1}^{1} \sum_{j=-1}^{1} T_t^{(i, j)}\right)\mathbf{1}' = \mathbf{1}'$, for all $(t = 1, 2, \ldots, n)$, we obtain

$$\mathbf{E}|_{u=1, v=1} \mathbf{1}' = 0.$$

Because each term in $\mathbf{E}|_{u=1, v=1}$ is nonnegative, each term in $\mathbf{E}|_{u=1, v=1}$ is $\mathbf{0}$. Hence,

$$\mathbf{E} = \mathbf{0},$$

and we obtain

$$\phi_t(u, v) = \phi_{t-1}(u, v)\left(\sum_{i=-1}^{1} \sum_{j=-1}^{1} T_t^{(i, j)} u^i v^j\right).$$

References

[1] S. Aki and K. Hirano, Sooner and later waiting time problems for runs in Markov dependent bivariate trials, Ann. Inst. Stat. Math. 51 (1999), 17-29.

[2] N. Balakrishnan, S. G. Mohanty and S. Aki, Start-up demonstration tests under Markov dependence model with corrective actions, Ann. Inst. Stat. Math. 49 (1997), 155-169.

[3] G. J. Chang, L. Cui and F. K. Hwang, Reliabilities of Consecutive-k System, Kluwer Academic Publishers, Netherlands, 2000.

[4] M. T. Chao, J. C. Fu and M. V. Koutras, Survey of reliability studies of consecutive-k-out-of-n: F and related systems, IEEE Trans. Reliability 40 (1995), 120-127.

[5] M. Doi and E. Yamamoto, On the joint distribution of runs in a sequence of multi-state trials, Stat. Probab. Lett. 39 (1998), 133-141.

[6] W. J. Ewens and G. R. Grant, Statistical Methods in Bioinformatics, Springer-Verlag, New York, 2001.

[7] W. Feller, An Introduction to Probability Theory and its Applications, Vol. I, 3rd ed., Wiley, New York, 1968.

[8] J. C. Fu and M. V. Koutras, Distribution theory of runs: a Markov chain approach, J. Amer. Stat. Assoc. 89 (1994), 1050-1058.

[9] J. C. Fu, Distribution theory of runs and patterns associated with a sequence of multi-state trials, Statistica Sinica 6 (1996), 957-974.

[10] A. P. Godbole, S. G. Papastavridi and R. S. Weishaar, Formulae and recursions for the joint distribution of success runs of several lengths, Ann. Inst. Stat. Math. 49 (1997), 141-153.

[11] Q. Han and S. Aki, Formulae and recursions for the joint distributions of success runs of several lengths in a two-state Markov chain, Stat. Probab. Lett. 40 (1998), 203-214.

[12] Q. Han and S. Aki, Joint distributions of runs in a sequence of multi-state trials, Ann. Inst. Stat. Math. 51 (1999), 419-447.

[13] Q. Han and S. Aki, Waiting time problems in a two-state Markov chain, Ann. Inst. Stat. Math. 52 (2000), 778-789.

[14] Q. Han, Joint distributions of success runs in a sequence of bivariate trials, Journal of East China Normal University (2002), to appear.

[15] M. V. Koutras and V. A. Alexandrou, Runs, scans and urn model distributions: a unified Markov chain approach, Ann. Inst. Stat. Math. 47 (1995), 743-766.

[16] M. V. Koutras and V. A. Alexandrou, Sooner waiting time problems in a sequence of trinomial trials, J. Appl. Prob. 34 (1997), 593-609.

[17] K. D. Ling, On binomial distributions of order k, Stat. Probab. Lett. 6 (1988), 247-250.

[18] A. M. Mood, The distribution theory of runs, Ann. Math. Stat. 11 (1940), 367-392.

■